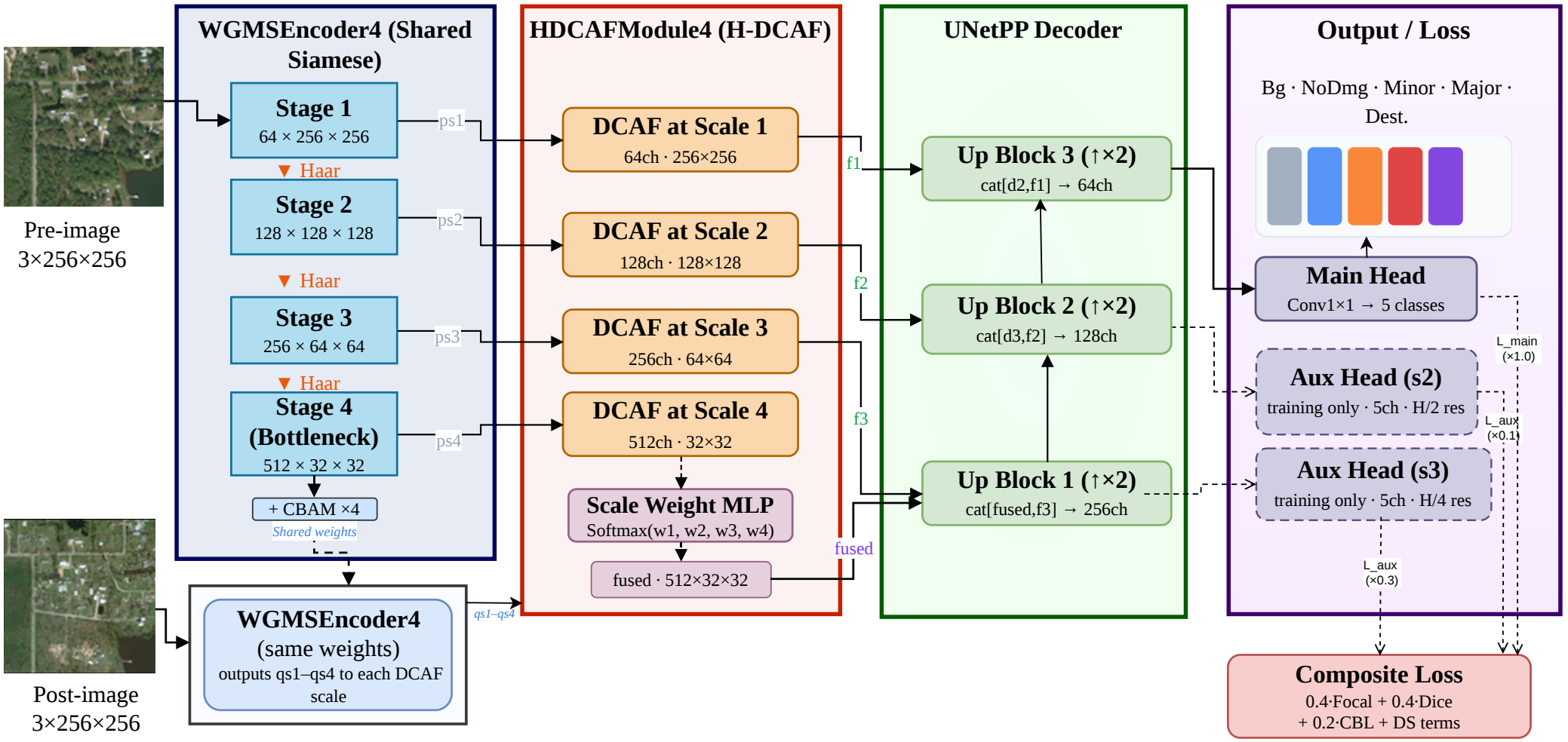

Pre-image
3×256×256
Post-image
3×256×256
WGMSEncoder4 (Shared Siamese)
Stage 1
64 × 256 × 256
▼ Haar
Stage 2
128 × 128 × 128
▼ Haar
Stage 3
256 × 64 × 64
▼ Haar
Stage 4 (Bottleneck)
512 × 32 × 32
+ CBAM ×4
Shared weights
ps1
ps2
ps3
ps4
WGMSEncoder4
(same weights)
outputs qs1–qs4 to each DCAF scale
qs1–qs4
HDCAFModule4 (H-DCAF)
DCAF at Scale 1
64ch · 256×256
DCAF at Scale 2
128ch · 128×128
DCAF at Scale 3
256ch · 64×64
DCAF at Scale 4
512ch · 32×32
Scale Weight MLP
Softmax(w1, w2, w3, w4)
fused · 512×32×32
f1
f2
f3
fused
UNetPP Decoder
Up Block 3 (↑×2)
cat[d2,f1] → 64ch
Up Block 2 (↑×2)
cat[d3,f2] → 128ch
Up Block 1 (↑×2)
cat[fused,f3] → 256ch
Output / Loss
Bg · NoDmg · Minor · Major · Dest.
Main Head
Conv1×1 → 5 classes
Aux Head (s2)
training only · 5ch · H/2 res
Aux Head (s3)
training only · 5ch · H/4 res
L_main (×1.0)
L_aux (×0.1)
L_aux (×0.3)
Composite Loss
0.4·Focal + 0.4·Dice
+ 0.2·CBL + DS terms
DCAFModule - Internal Flow (one scale)
pre feat
post feat
|diff|
query q
gate mask
fused feat
Conv 3×3
→ output
α·pre+β·post
Legend
Encoder Feature Map
DCAF Fusion
Decoder
Output Head
Aux Head (train only)
Composite Loss